

Developing generative AI solutions with foundation models

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You can develop generative AI solutions with foundation models in IBM watsonx.ai. You can create prompts to generate, classify, summarize, or extract content from your input content. Choose from IBM models or open source models. You can tune foundation models to customize your prompt output or optimize inferencing performance.

Generative AI capabilities

With watsonx.ai, you can create generative AI solutions that include the following capabilities and resources.

Prompting

Build prompts that instruct a foundation model to generate a response. You can chat with documents and other media, include variables for reusing prompts, remove harmful content, and control other prompt and model settings.

See [Building prompts](#).

Agentic AI

Build workflows for the foundation model to function as an agent that controls the flow of interaction with the user.

See [Building agent-driven workflows with the Chat API](#).

Retrieval-augmented generation

Build a RAG pattern to ground the model in facts from your documents. You can customize your RAG pattern to extract text from documents, create vector indexes, and rerank retrieved content. You can automate the search for an optimized, production-quality, RAG pattern based on your data and use-case.

See [Retrieval-augmented generation](#).

Forecasting

Forecast future values based on historic time series data.

See [Forecasting future values](#).

Foundation models

Watsonx.ai has a range of foundation models that you can try. Otherwise, you can select a deploy-on-demand model or import and deploy a custom model.

See [Supported foundation models](#).

Model tuning

Tune a smaller foundation model to achieve results comparable to larger models in the same model family.

See [Tuning foundation models](#).

Evaluations

Evaluate and compare your generative AI assets with quantitative metrics and customizable criteria that fit your use cases.

See [Evaluating AI models](#).

For tips on planning your generative AI solution, see [Planning an AI solution](#).

Ways to work [↗](#)

You can write code for your generative AI solution. For a list of tasks and APIs, see [Coding generative AI solutions](#).

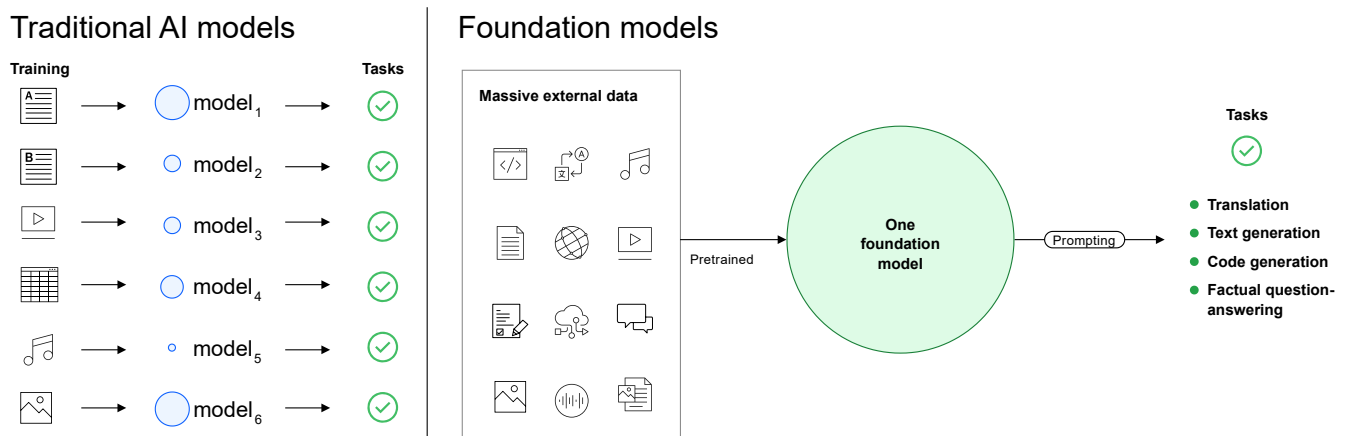
You can work in a no-code or low-code experience with the following tools in the UI:

- [Prompt Lab](#)
- [Agent Lab](#)
- [AutoAI for RAG](#)
- [Tuning Studio](#)
- [Evaluation Studio](#)
- [Vector indexes](#)

Foundation model architecture [↗](#)

Foundation models are large AI models that have billions of parameters and are trained on terabytes of data. Foundation models can do various tasks, including text, code, or image generation, classification, conversation, and more. Large language models are a subset of foundation models that can do tasks that are related to text and code.

Foundation models represent a fundamentally different model architecture and purpose for AI systems. The following diagram illustrates the difference between traditional machine learning AI models and foundation models for generative AI.



As shown in the diagram, traditional AI models specialize in specific tasks. Most traditional AI models are built by using machine learning, which requires a large, structured, well-labeled data set that encompasses a specific task that you want to tackle. Often these datasets must be sourced, curated, and labeled by hand, a job that requires people with domain knowledge and takes time. After it is trained, a traditional AI model can do a single task well. The traditional AI model uses what it learns from patterns in the training data to predict outcomes in unknown data. You can create machine learning models for your specific use cases with tools like AutoAI and Jupyter notebooks, and then deploy them.

In contrast, foundation models are trained on large, diverse, unlabeled datasets and can be used for many different tasks. Foundation models were first used to generate text by calculating the most-probable next word in natural language translation tasks. However, model providers are learning that, when prompted with the right input, foundation models can do various other tasks well. Instead of creating your own foundation models, you use existing deployed models and engineer prompts to generate the results that you need.

Learn more [↗](#)

- [Prompt Lab](#)
- [Tuning Studio](#)
- [Agent Lab \(beta\)](#)
- [Coding generative AI solutions](#)
- [Security and privacy](#)
- [Model terms of use](#)
- [Tokens](#)
- [Retrieval-augmented generation](#)
- [AI risk atlas](#)
- [Quick start tutorials: Working with generative AI](#)